

Manuel Magana

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EDUCATION

Northeastern University

M.S. Computer Science

San Jose, CA

May 2026

University of California, Berkeley

B.A. Molecular and Cell Biology: Neurobiology

Berkeley, CA

May 2021

EXPERIENCE

Genentech

Bioinformatics Scientist

San Francisco, CA

Nov 2022 – Jul 2025

- Designed a data validation library in R, then collaborated with 2 computational scientists to port it to Python with branching strategies, tests, and PR reviews, reducing manual data review from days to hours.
- Built a PySpark pipeline to filter, clean, and visualize proteins of interest from a 200GB+ mass spectrometry dataset, unblocking a study that exceeded the team's existing R-based tooling capacity.
- Implemented features in an R Shiny dashboard with one other scientist, extending support beyond mass spec to additional data types so 4 team scientists could reliably repeat the same analysis across different datasets.
- Containerized analytical tools with Docker for integration into an internal workflow system used by 20+ scientists, eliminating environment setup issues and enabling reproducible, one-command execution.

Endless West

Junior Data Scientist

San Francisco, CA

Mar 2021 – Nov 2021

- Built ML training pipelines (Logistic Regression, KNN, XGBoost, Random Forest) for sensory attribute prediction from chromatography data, enabling R&D to prioritize synthesis targets by predicted flavor profile instead of trial and error.
- Developed Python web scrapers using BeautifulSoup to extract and structure chemical property data from public databases, replacing hours of manual lookup per compound with a searchable local dataset.

PROJECTS

WhatToDo | AI Event Discovery Platform | *Python, FastAPI, Docker, AWS* | [GitHub](#) *Jan 2026 – Present*

- Built a dual-provider AI recommendation engine with JWT auth, SQLAlchemy/PostgreSQL on AWS RDS, and deployed via App Runner with GitHub Actions CI/CD, giving users personalized event suggestions from two LLM providers with automatic failover.
- Led backend development for a 4-person team; achieved 96% pytest coverage across auth, AI services, and the event planner, ensuring safe refactors as the API surface grew.

Physiological Signal Analytics Platform | *Python* | [API](#) | [Library](#)

Sept 2025 – Jan 2026

- Architected a two-layer system: a Python ETL/analysis library for batch processing and a production REST API, separating offline analysis from real-time access so researchers and front-end clients could query independently.
- Built a FastAPI service over DuckDB with offset pagination, composable query filters, and Pydantic validation, enabling efficient retrieval of raw signals and derived features without loading full datasets into memory.
- Implemented analysis endpoints for PCA, FFT, and spectral density, letting researchers interactively explore physiological signals without writing custom scripts.

TECHNICAL SKILLS

Languages: Python, R, Go, Java, SQL, Bash

Frameworks, Libraries, and Tools: FastAPI, SpringBoot, pytest, JUnit, AWS, Git, Redis, SQL, pandas, PySpark, scikit-learn, PyTorch, TensorFlow/Keras, Docker, GitHub Actions, CI/CD